FITNESS CENTER MANAGEMENT SYSTEM

Phase 2: Backend Development & Configurations

Internship Evaluation Project Documentation

Organization: FitHub Fitness Center

Platform: Salesforce CRM

PROJECT TEAM DETAILS

S. No	Name	Role	Phase Responsibility
1	Aakash U	Team Lead / Developer	Phase 1 – Requirement Analysis & Planning
2	Saiprasanth K	Developer	Phase 2 – Backend Development & Configurations
3	Sanjay E	UI/UX Designer	Phase 3 – UI/UX Development & Customization
4	Abilash P	Tester / Analyst	Phase 4 – Testing & Data Handling

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1. PHASE OVERVIEW

This phase focuses on developing the backend architecture of the FitHub CRM system using Salesforce. It includes designing the data model, configuring objects and fields, implementing automation using flows and validation rules, and developing Apex logic to support complex business processes.

2. SALESFORCE ACCOUNT SETUP

Objective

To establish a Salesforce development environment for building the application.

Description

Salesforce is a cloud-based Customer Relationship Management (CRM) platform that enables organizations to manage customer data, automate workflows, and enhance operational efficiency.

Tasks Completed

- Created Salesforce Developer Edition account
- Accessed Setup and Developer Console
- Explored core features and system navigation

3. CUSTOM OBJECTS CREATION

Objective

To define the core data structure of the application.

Objects Created

- Member
- Trainer
- Class
- Booking

Configuration Details

- Enabled reporting and search functionality
- Configured record naming using Auto Number format
- Established relationships between objects

4. TAB CREATION

Objective

To provide easy navigation within the application.

Tasks Completed

Tabs were created for:

- Member
- Trainer
- Class

- Booking

Outcome

Users can efficiently navigate and access all objects directly from the application interface.

5. FIELDS CONFIGURATION

Objective

To define detailed attributes for each object.

Member Object

- Member ID (Auto Number)
- Full Name (Text)
- Membership Status (Formula)
- Email (Email)
- Phone Number (Phone)
- Membership Type (Picklist)
- Subscription Start Date (Date)
- Subscription End Date (Date)

Trainer Object

- Trainer ID (Auto Number)
- Full Name (Text)
- Specialization (Picklist)
- Availability (Checkbox)
- Salary (Currency)

Class Object

- Class Name (Text)
- Class Type (Picklist)
- Trainer (Lookup)
- Schedule (Date/Time)
- Capacity (Number)
- Booked Slots (Roll-Up Summary)
- Status (Formula)

Booking Object

- Booking ID (Auto Number)
- Booking Type (Picklist)
- Member (Master-Detail)
- Class (Master-Detail)
- Amount (Currency)
- Payment Status (Picklist)

6. VALIDATION RULES

Objective

To ensure data accuracy and enforce business constraints.

Rules Implemented

Prevent Overbooking

- Condition: Booked Slots ≥ Capacity

- Error Message: *"You cannot book more slots than the class capacity."*

Payment Confirmation Rule

- Condition: Booking confirmed without payment
- Error Message: *"Booking cannot be confirmed until payment is marked as Paid."*

7. FLOWS

Objective

To automate business processes without writing code.

Flow Implemented

Membership Expiry Notification Flow

Functionality

- Trigger: Member record update
- Condition: Membership Status = Expired
- Action: Sends email notification

Outcome

Ensures timely communication with members regarding expired memberships.

8. APEX TRIGGERS

Objective

To implement advanced business logic beyond declarative tools.

Trigger 1: Class Trigger

Purpose: Automatically assign trainers

Logic:

- Assign trainer based on class type and specialization
- Check trainer availability
- Prevent schedule conflicts
- Ignore roll-up summary field updates

Trigger 2: Booking Trigger

Purpose: Automate booking calculations and updates

Logic:

- Calculate booking amount based on:
  - Membership Type
  - Class Type
- Update:
  - Subscription Start Date
  - Subscription End Date

9. SCHEDULEABLE APEX

Objective

To automate scheduled operations.

Implementation

Subscription Reminder Scheduler

Functionality

- Executes daily at 6 AM
- Identifies members with subscriptions expiring in 5 days
- Sends automated email reminders

Outcome

Improves member retention through proactive communication.

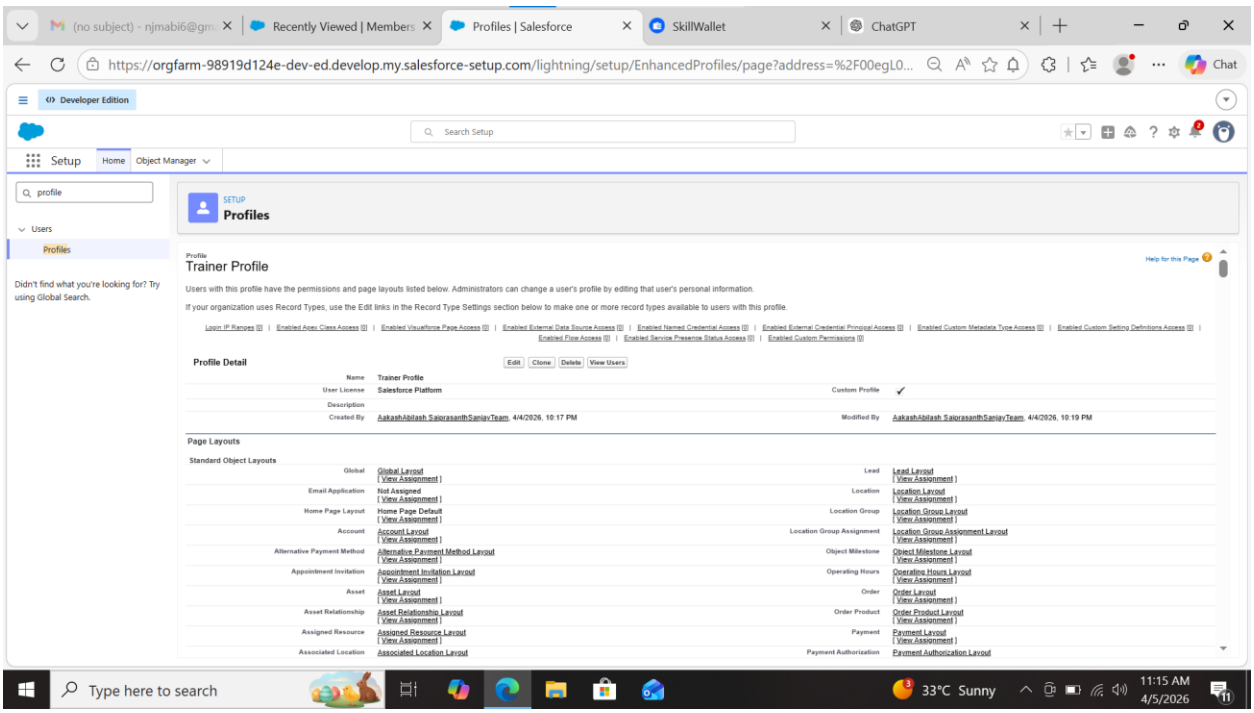
10. TESTING & VALIDATION

Activities Performed

- Verified object relationships
- Tested validation rules
- Executed flows and Apex triggers
- Validated email notifications

11. CHALLENGES FACED

- Managing trigger execution conditions
- Handling complex object relationships
- Preventing scheduling conflicts in trainer assignments

12. CONCLUSION

Phase 2 successfully establishes the backend foundation of the FitHub CRM system. With properly configured data structures, automation tools, and Apex logic, the system efficiently handles core business operations while ensuring accuracy, scalability, and reliability.